Assignment 2 Script

Christina Treacy

2026-05-07

```r
library(dataRetrieval)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(here)
```

```
here() starts at /Users/christinatreacy/Desktop/course stuff/spring 26/ESM 262/Assignment_2
```

```r
library(usethis)
library(lubridate)
library(broom)
```

Load in data from NWIS server and save to CSV

```r
siteNumbers <- c("14211720", "14211010", "14166000")
parameterCd <- "00010" # water temperature
startDate <- "2010-01-01"
endDate <- "2023-12-31"
temp_data <- readNWISdv(siteNumbers, parameterCd, startDate, endDate)
```

```
Warning in readNWISdv(siteNumbers, parameterCd, startDate, endDate): NWIS
servers are slated for decommission. Please begin to migrate to
read_waterdata_daily.
```

```
GET: https://waterservices.usgs.gov/nwis/dv/?site=14211720%2C14211010%2C14166000&format=wate
01-01&endDT=2023-12-31
```

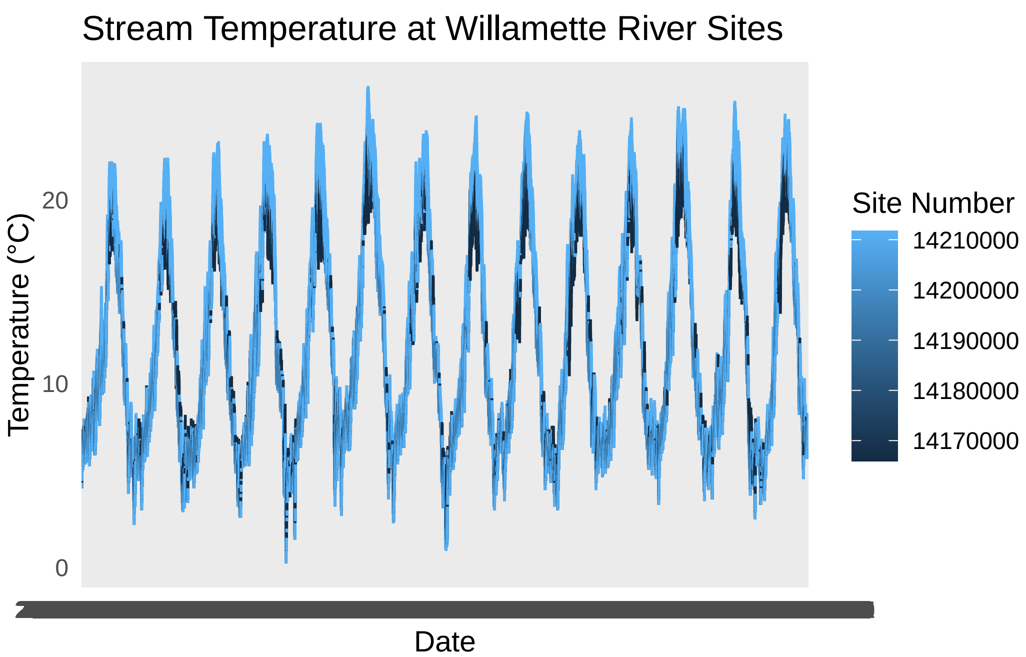
```
# rename columns for clarity
colnames(temp_data) <- c("agency_cd", "site_no", "Date", "temp_C")
# extract some date info for use later
temp_data$month = month(temp_data$Date, label=TRUE)
temp_data$year = year(temp_data$Date)

write.csv(temp_data, "tempdata.csv", row.names = FALSE)
```

Load data directly from repository

```
tempdata <- read.csv(here("tempdata.csv"))
```

Plot data & Calculate Summary Statistics

```
# check by plotting
ggplot(tempdata, aes(x=Date, y=temp_C, color=site_no)) +
geom_line() +
labs(title="Stream Temperature at Willamette River Sites",
x="Date", y="Temperature (°C)", color="Site Number") +
theme_minimal()
```




```
# generate summary stats
summary_table <- tempdata |>
  group_by(site_no) |>
  summarize(min=min(temp_C),
            mean=mean(temp_C),
            med=median(temp_C),
            max=max(temp_C),
            st_dev=sd(temp_C))
write.csv(summary_table, here("site_temp_summary_table.csv"))
```

Trends in Water Temperature for the month of August

```
# filter for August temperatures
aug_data <- subset(temp_data, month == "Aug")

# fit a GLM model to see if there is a trend in August temperatures over time, separated by s
aug_model <- glm(temp_C ~ year*site_no, data = aug_data)
summary(aug_model)
```

```
Call:
glm(formula = temp_C ~ year * site_no, data = aug_data)
```

```
Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)         -243.03103   25.50634  -9.528  < 2e-16 ***
year                   0.12975    0.01265  10.258  < 2e-16 ***
site_no14211010      -91.98714   36.01709  -2.554  0.01076 *
site_no14211720      -28.41636   36.08229  -0.788  0.43111
year:site_no14211010   0.04627    0.01786   2.590  0.00969 **
year:site_no14211720   0.01619    0.01789   0.905  0.36581
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 1.121551)

    Null deviance: 5985.3  on 1298  degrees of freedom
Residual deviance: 1450.2  on 1293  degrees of freedom
AIC: 3843.4

Number of Fisher Scoring iterations: 2
```

The slope for first site is 0.12975, showing a slight positive trend in water temperature.

Site 14211010 has a slope that is 0.04627 greater than the first site, but the p-value is greater than 0.05, meaning that this result is not statistically significant. Site 14211720 has a slope that is 0.01619 greater than the first site. Both of these results demonstrate relatively small differences in trends between sites.

Trends in Water Temperature for the month of February

```r
# filter for February temperatures
feb_data <- subset(temp_data, month == "Feb")

# fit a GLM model to see if there is a trend in Feburary temperatures over time, separated by
feb_model <- glm(temp_C ~ year*site_no, data = feb_data)
summary(feb_model)
```

```
Call:
glm(formula = temp_C ~ year * site_no, data = feb_data)

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)         136.647423  28.517563   4.792 1.86e-06 ***
```

```
year                   -0.064277   0.014142  -4.545 6.06e-06 ***
site_no14211010      -40.061916  40.387842  -0.992    0.321
site_no14211720       -8.440431  40.329925  -0.209    0.834
year:site_no14211010   0.019204   0.020029   0.959    0.338
year:site_no14211720   0.004215   0.020000   0.211    0.833
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 1.280544)

    Null deviance: 2060.3  on 1182  degrees of freedom
Residual deviance: 1507.2  on 1177  degrees of freedom
AIC: 3657.7

Number of Fisher Scoring iterations: 2
```

The slope for first site is -0.064277, showing a slight negative trend in water temperature over time in the month of February.

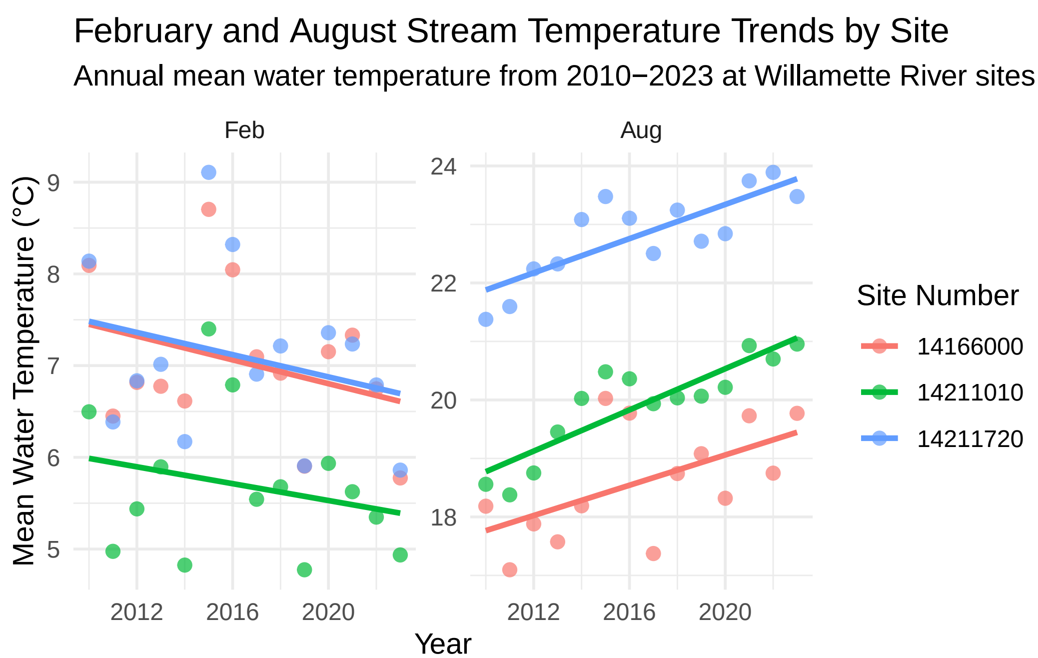
Site 14211010 has a slope that is 0.019204 greater than the first site, and Site 14211720 has a slope that is 0.01619 greater than the first site. Both of these results demonstrate relatively small differences in trends between sites.However, the p-value is greater than 0.05 for both of these estimates, meaning that these results are not statistically significant.

Plot Combined Analysis

```
# Load saved data from repository
tempdata <- read.csv(here("tempdata.csv"))

# Clean date variables after loading from CSV
tempdata <- tempdata |>
  mutate(
    Date = as.Date(Date),
    site_no = as.factor(site_no),
    month = month(Date, label = TRUE, abbr = TRUE),
    year = year(Date)
  )

# Keep only the two months analyzed
monthly_data <- tempdata |>
  filter(month %in% c("Feb", "Aug"))

# Calculate annual average temperature for each month and site
```

```r
monthly_summary <- monthly_data |>
  group_by(month, year, site_no) |>
  summarize(
    mean_temp_C = mean(temp_C, na.rm = TRUE),
    .groups = "drop"
  )

# Summary plot combining February and August trends
ggplot(monthly_summary, aes(x = year, y = mean_temp_C, color = site_no)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1) +
  facet_wrap(~ month, scales = "free_y") +
  labs(
    title = "February and August Stream Temperature Trends by Site",
    subtitle = "Annual mean water temperature from 2010-2023 at Willamette River sites",
    x = "Year",
    y = "Mean Water Temperature (°C)",
    color = "Site Number"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Combined Analysis Conclusions

The combined monthly analysis suggests that stream temperature trends differ by season. August temperatures show a slight positive trend over time, indicating that late-summer water temperatures may be increasing across the Willamette River sites. This pattern is meaningful because August is typically a warmer month, so even small increases in stream temperature could matter for aquatic ecosystems and water-quality management. In contrast, February temperatures show a slight negative trend or relatively weak change over time, suggesting that winter stream temperatures are not warming in the same way as summer temperatures. The differences between sites appear relatively small, and several site-specific interaction terms were not statistically significant, meaning there is limited evidence that the sites are experiencing strongly different trends. Overall, the most important management takeaway is that warming appears more evident in August than in February, which suggests that late-summer stream conditions may deserve closer monitoring.